Re-purposing survey data sources for teaching and learning

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Overview of talk

- funding initiative for 're-purposing' digital content: Exchange for Learning Programme (X4L)
- rationale and objectives of our X4L project: Exploiting Survey Data Sources in the Classroom
- anticipated national infrastructure for preserving/accessing teaching and learning (T&L) materials
- metadata standards for T&L materials
- most effective ways of collaborating...

Exchange for Learning (X4L) Programme

- funded by the Joint Information Systems Committee (JISC) with support from other T&L bodies
- motivated by the need to make the most of the considerable investment in a range of digital content which has high potential value for use in T&L
- explore ways of re-purposing existing and forthcoming JISC funded content + other content (where IPR can be negotiated)
- contribute to development of mechanisms and supporting services to allow the submission and sharing of content

Aims and Challenges

- encourage HE/FE institutions to actively take a role in defining the kinds of content that would best fit their learning aims
- encourage national agencies and support services to work to provide the tools and infrastructure to allow this "exchange of learning" to take place
- challenge is to demonstrate how existing content can be re-used to support learning
- pedagogical outcomes at the heart of the programme - focus on learning activities and outcomes
- explore sustainability and widespread adoption of e-learning materials - by unlocking the potential of complementary areas of work

Strand A - content

- assemble "chunks" of content which can be identified as learning objects to create new learning materials
- make more direct connections between available resources and the 16+ curriculum and to create learning pathways
- document the processes used to re-purpose learning materials and to provide case studies and exemplars
- test the usability of materials within the context of an institutional environment/portal or within a local VLE
- submit these learning materials for sharing to suitable learning materials repository
- tag/catalogue the 'learning objects' using appropriate metadata
- to trial and evaluate their use within the community

Strand B: Tools and "Service" Architecture

- development of tools and management of records and processes leading to the creation of an architecture to support the programme
- development of tools based upon appropriate standards and specifications to facilitate the creation of learning pathways and the assembly of learning objects and appropriate metadata
- create a development bay for learning objects for depositing/archiving materials with a view to longer-term use within a service environment

Application restrictions

- proposals invited both from single institutions and consortia
- partnerships actively encouraged with owners or suppliers of relevant content
- JISC-funded services invited in partnership with colleges and universities
- FE colleges high on the agenda
- consider projects 1-3 years duration; budgets ranging from £10,000 to £300,000

Immediate Problem

- not a widely established history of JISC service providers working closely with FE institutions
- some instances - EDINA/MIMAS with NLN
- UK Data Archive
  - few active links with FE (16-18)
  - no dedicated resources/staff expertise to support this community explicitly
  - occasional seminars run largely about resource discovery and data potential
- hence many ‘consortia’ formed overnight
- no time to ‘get to know each other’ well enough

More problems

- Project activity should not duplicate the effort that has gone in to previous JISC initiatives:

— DNER Learning and Teaching Programme

  — Not finished at start of programme, so many projects not completed

— Teaching and Learning Technology Programme (TLTP)

  — TLTP - a multi million £ exercise in 1990s with little spin off or use of the resulting ideas/tools

UK Data Archive Project

- UKDA has rich stock of data sources that we would like to encourage the use of in T&L
- formed a consortia …
  - University of Essex Government Department
  - local FE college Social Science department
- project focus on teaching politics at undergraduate and A' level (16-18 academic qualification) levels

Exploiting UK survey data sources for teaching political science: experiences from the classroom

Project rationale

- UK academic community has access to a unique and expansive range of digital data resources
- whilst individual datasets used extensively in academic research they are significantly under used in T&L programmes within HE, and rarely used in Further Education
- UKDA has the potential to offer its resources to the T&L communities for developing more 'packaged' resources BUT needs the advice and input from instructors in the classroom on how to re-purpose and apply the content
- widely recognised in the UK that the skills shortage of quantitative analysts is now critical - introducing concepts early on in post-16 education is one way to redress this shortage

Project objectives

- produce teaching datasets from more complex socio-economic datasets
- develop integrated web-based T&L resources that link together resource discovery tools, and data exploration and extraction tools (NESSTAR) with teaching materials that help address substantive issues for political science teaching
- gain evaluation and feedback from piloting this re-purposed content - political science teachers and students
- suggest models for improving the productivity of teachers by reducing the resources required to incorporate data related resources into T&L
- promote increased and more effective use of a national data services for problem-based learning in the classroom

Course aspects covered

- Substantive areas: national issues/controversies in politics
  - — voting behaviour
  - — crime and social order
  - — race and immigration

- Empirical orientations: research methods in politics
  - — potential of survey data to answer questions
  - — survey measurement; sampling
  - — basic data management/basic data analysis
  - — resource discovery skills

- Essex HE course already combines these components
- Local college course does not 'use' data; no hands on data manipulation; no aim to improve statistical literacy

Datasets selected

- Plan to use British Election Studies
- BUT FE syllabus dropped voter behaviour module SO using British Crime Survey and British Social Attitudes Survey
- Create mini cut-down teaching datasets for BCS and BSA - publish in NESSTAR

Materials to be created

- Web-based learners' and teachers' guides - tutorials and exercises
- User guides to aid resource discovery, web-based data extraction and data exploration/visualisation of the teaching datsets via the NESSTAR software suite, suitable for student use
- Enhancement of the teaching datasets and learning text/exercises through adding and linking other relevant information sources
- Metadata of the learning objects for submission to the Repository
- A fully documented 'warts and all' report on the processes used to re-purpose and pilot the learning materials

Metadata

- standards changing rapidly through X4L programme
- repository/delivery - JORUM and RELOAD projects
- draft X4L Application Profile developed by the UK Metadata for Education Group (to identify common UK practice in the use of metadata in packaged e-learning content)
- **UK Common Metadata Framework** (UKCMF) developed - an application profile of the IEEE Learning Object Metadata Standard (LOM) that has been optimised for use within the context of UK education
- IEEE = US Institute of Electrical and Electronics Engineers
- UKCMF heavily influenced by the work of Canadian Core Learning Object Metadata Application Profile (CanCor)

JORUM

- MIMAS/EDINA
- provide a practical test bed investigation into a digital repository for learning materials
- able to provide technical infrastructure for storage and retrieval of learning objects
- functionality extends to disaggregation; re-purposing and re-aggregation of learning objects
- investigate requirements for a long-term digital repository service

Future steps

- Advocate collaboration with other e-learning projects
- Recommend sharing of modules/components instead of re-investing the wheel
- Share best stories/practice
- Facilitate collaborative working via the IASSIST Education Committee